



INSTALLATION, OPERATION AND
MAINTENANCE MANUAL FOR CENTRIFUGAL
PUMPS
TYPE **NHVf**

Original Manual

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1. GENERAL

This manual must be read and understood by persons responsible for installation and operation of the pump before it is installed and operated. Operators are responsible for following the provisions of this manual during each pump operation stage.

The manufacturer „Hydro-Vacuum” S.A. does not take any responsibility for if the pump is operated in conditions other than conditions approved by the manufacturer in writing.

This guarantee does not apply if the provisions of this manual are breached, in particular in case the device is operated outside its intended use or in an incorrect manner or the pump is dismantled in an unauthorised manner. The warranty does not cover natural wear and tear of the pump and its wear parts, i.e.: impeller, seals, shaft, protective sleeves, bearings, slip rings, etc. or damage occurred during shipment.

1.1. Pump marking

The pump data plate includes information regarding pump type, size, constructional and material executions as well as nominal operating parameters. The data plate marking must be cited in case of contacts with the manufacturer.



Fig.1.1 NHVf series pump data plate with Ex sign

Type – Indicates pump type and size including material and design version

Q – Indicates efficiency in an operating point or pump operation range,

H – Indicates delivery head in the operating point or operation range,

Nr – Indicates pump serial number

Rok – Indicates year of manufacture

Ex – Indicates pump for use in areas at risk of explosion

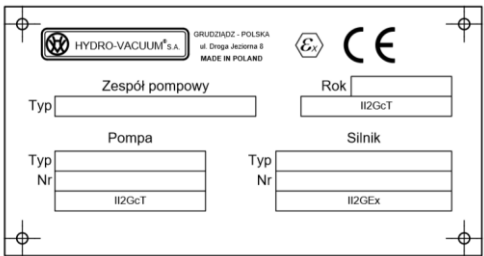


Fig.1.2 NHVf series pump unit data plate with Ex sign

2. SAFETY

2.1. Symbols used in the manual

The manual contains the following symbols regarding safety of operation and device:



General danger symbol specifically regarding operation of the pumping unit – failure to comply with these warnings may result in endangering personnel's life or health!

ATTENTION!

Safety note - in particular regarding correct pump operation - failure to comply with these warnings may lead to damage of the device!



A warning regarding a voltage presence – failure to comply with these warnings may result in an electric shock or damage.



Marking of explosion protection

Symbol indicates information on explosion protection in areas with danger of explosions, in accordance with 2014/34 UE (ATEX).

2.2. General informations

NHVf series pumps comply with the requirements of Machine Directive 2006/42/EG.



The personnel employed to install, operate, inspect and maintain the device, must prove they have sufficient knowledge on safety regulations, have all required qualifications as well as know and comply with instructions provided in this manual.

ATTENTION!

The pump unit must not be operated in a manner contrary to its intended use as regards physical-chemical properties of the pumped medium, i.e. capacity, pressure, temperature, density, aggressiveness, abrasion, rotational speed direction and value as well as other parameters defined in pump technical data or offer details. If a pump is to be used for a different application, contact a sales office or manufacturer's representative. Hydro-Vacuum S.A. shall not be responsible in case a pump is used for other purposes without its prior explicit consent provided in writing.

ATTENTION!	The pump unit must not be operated within an installation constructed contrary to the art of engineering, in particular as regards the design of the suction and delivery line and ensuring the permissible operating field and temperature.
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ATTENTION!	The operator is responsible for maintaining the pump unit in a proper technical condition, prompt inspections and replacement of worn-out parts. It is forbidden to activate the pumping engine when it is dismantled or not connected to the installation.
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2.3. Safety regulations

The manual contains basic requirements that must be complied with during installation, operation, inspections and repairs of the pump and pumping unit.

The pump operating manual must always be located in a visible and accessible location in the operation area. In case the pump or pumping unit is moved or transferred to a third person, the manual must be delivered to the end-user.

This manual does not refer to general accident prevention regulations and does not take into account local industrial safety and legal regulations that must be applied.

This manual does not include information regarding protective devices and methods of recycling of the pumped medium and/or media used to rinse or lubricate bearings, in particular as regards their toxicity. Personnel is responsible for correct operation of the pumping motor, in line with applicable regulations.

2.4. Consequences and threats resulting from failure to comply with safety instructions

- Failure to comply with this operating manual results in loss of rights arising out of the warranty and liability for damages.
- Failure to comply with this manual may lead to the following risks:



Risks for personnel resulting from impact exerted by an electric field, magnetic field, high temperatures, mechanical and chemical impact as well as explosion.

Risk for environment from leakage of dangerous substances.

ATTENTION!	Failure of important product functions. Failure of measuring devices and electronic equipment resulting from magnetic field impact.
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Risk of explosion especially for products for use in hazardous areas. Products for use in hazardous areas must be compatible and properly marked. It is recommended to observe relevant instructions

2.5. Personnel safety



Failure to comply with any of the following provisions results in risks to personnel's health or life and property damage.

- Usage of product components depends on individual operating conditions thus the operator must ensure regular inspections and maintenance. In case any irregularities regarding the product condition or abnormal operating parameters are noticed, the pump must be immediately stopped. Otherwise, the product may be damaged and, in particular, personnel safety is compromised.
- Personnel must have and use appropriate personal protective equipment.
- As regards systems or products whose damage may result in endangering personnel's safety or life, it must be equipped with alarm devices which are constantly in the standby mode.
- In case of systems or products used to pump low or high temperature liquids which may cause injuries, these systems must be protected against accidental contact and marked as applicable
- Moving elements of the pump or pump unit must be protected against accidental contact by providing necessary guards. Do not disassemble moving element guards during device operation. An exception from this rule is the procedure of adjusting the packing gland which must be conducted by qualified personnel. After the procedure is completed, guards must always be reassembled.
- In case of pumps or systems whose noise emission level exceeds 85 dB, personnel must use applicable hearing protection measures.
- Leakage (e.g. sealings) dangerous (e.g. toxic agents, explosive, hot or cold substances) must be handled in a manner safe for personnel and the environment. Always observe all applicable legal regulations.



Threat connected with electricity must be eliminated on the basis of local regulations regarding electric equipment. During work conducted on electric equipment, it must be effectively isolated from the supply source by deactivating a switch and removing a fuse or permanently isolated and secured against accidental activation.

2.6. Maintenance, inspection and repair works safety



Failure to comply with any of the following provisions results in risks to personnel's health or life and property damage.

- Any work on the pumping unit must be conducted by authorised and qualified personnel familiar with the content of this manual.
- Any work on the pumping unit must be conducted during its standstill, in line with the following rules:
 - Secure the unit against unintentional power-up.
 - Internal pump pressure must be equal to the ambient pressure.
 - The pump should be emptied of pumped medium.
 - Temperature of unit elements must be equal to the ambient temperature.
 - In case dangerous substances are pumped, internal elements of the pump and auxiliary systems must be disinfected.
 - Immediately after work is completed, reassemble and activate all protective and safety devices in their mounting locations.

ATTENTION!

Failure to comply with any of the following provisions results in risks to correct device operation.

- Any work on a pumping unit still covered by a warranty must be conducted by an authorised manufacturer's service department or with its written consent. After the warranty period expires, it is recommended that any work, in particular repair works activities, should be conducted by an authorised manufacturer's service department..
- Any modification of the pump or pumping unit is permitted only subject to manufacturer's written consent.
- Parts used to conduct work on the unit must be genuine and approved by the manufacturer. Using unauthorised parts results in warranty becoming void and all related consequences.

2.7. Safety of use in hazardous areas



Operating pump unit in explosion hazardous areas is permissible in accordance with the operation and maintenance manual and the following instructions.



Products used in explosion hazardous areas must be compatible and properly marked.

Explosion protection is ensured if the product is used as intended only. The pump unit must not operate outside its performance limits.

2.7.1. Marking



Marking on the pump applies only to the specific pump.
The coupling and the motor should have separate markings and should be delivered with the relevant declarations of conformity.

Table 2.1 Example marking of an equipment for use in explosive atmospheres: II 2GhExII BT

	Symbol	Meaning
Explosion group	I	Device for use in underground mines and surface facilities susceptible to gas or dust explosion hazards.
	II	For use in all other industries in surface facilities susceptible to gas, vapour, dust and mist explosion hazard.
Equipment category	1	Device designed for use as intended by the manufacturer, providing very high level of protection, even for rarely occurring failures. In case of a failure of a single protective device, another one provides a required safety level or the safety level is provided for two independent failures.
	2	Device for use as intended by the manufacturer, providing high safety level, even for frequently occurring disturbances or failures.
	3	Equipment for use as intended by the manufacturer, providing normal safety level during operation in normal conditions.
Atmosphere	G	Gas
	P	Dust
Marking of explosion protection according to PN-EN ISO 80079-37	h	Under which there may be the following types of non-electrical protection:
		„c” constructional safety
		„b” control of ignition source
		„k” liquid immersion

Type NHVf centrifugal pumps are equipped with a type of non-electrical protection "c" - design safety

Pump's temperature class is determined based on the limit temperatures for a specific pump type (see below).

For pump units with components rated at different temperature classes, use in the lowest temperature class only.

2.7.2. Operating temperature



In normal conditions, the highest temperature may occur at the pump casing and the bearing mount surface, as well as the shaft packing and the external mechanical seal system, if used.

The temperature at the pump casing surface corresponds to the medium temperature.

For pumps with heating jacket, please observe the relevant system temperature class.

The temperature class presents a maximum temperature value at the pump unit surface permitted for use with medium at the maximum limit temperature. Permissible temperature class is displayed on the pump data plate. Detailed data on maximum permissible medium temperature appears in offer documentation or data sheet.

Table 2.2 Temperature class

<i>Temperature class to EN 13463-1</i>	<i>Maximum pump casing temperature</i>	<i>Maximum medium temperature</i>
T4	135°C	110°C
T3	200°C	140°C

For the rolling bearing area T4 temperature class is guaranteed provided that the ambient temperature does not exceed 40°C and the pump unit is used as intended.



During work protect the pump against excessive accumulation of sediments, which may cause the temperature to rise above the permitted value.

Provide free air flow through the bearing nodes to ensure the required heat exchange conditions.

2.7.3. Filling the pump and system



In order to prevent the explosive atmosphere from forming when transferring flammable media, make sure the pump and lines are filled with the medium at all times.

If the pump and the lines cannot be filled with medium at all times, use suitable monitoring (e.g. inlet pressure measurement, medium level sensor) to prevent dry run and temperature rise above the permissible limits.

2.7.4. Checking motor direction

Motor should rotate clockwise as viewed from the motor side. It is recommended check motor direction with coupling halves decoupled. Otherwise, check motor direction by running the pump unit connected to the system and filled with medium for several seconds.



It is prohibited run the pump unit without connecting it to the system and filling with medium. Dry run may lead to a rise in temperature above its permissible limit, explosion and damage to pump components.

2.7.5. Pump start-up and operation

Pump can be started after all conditions in section 6 are met, and all precautions are taken for the pumps for use in hazardous areas.



Before start-up, fully open the suction line valve and open the discharge line valve approximately 25%. Immediately after start-up, set the valve to enable operation within its performance limits. Pump can also be started with the discharge line valve open if other adjustment methods are used, i.e. frequency inverter.



It is prohibited to run the pump with suction and/or discharge line valve closed to prevent rapid rise in temperature.



For pumps with mechanical seals, the temperature may rise above its limit value near the shaft packing due to the dry run conditions. Dry run may be due to:

- incorrect installation and running without medium,
- transferring medium with high gas content,
- pump operation outside its performance limits,
- incorrect suction line configuration.

2.7.6. Apparatus and additional devices



It is prohibited to install electrical devices, control devices and accessories (e.g. mechanical seal system) which does not meet all relevant safety and explosion protection requirements.

2.7.7. Maintenance



Ensure correct maintenance and inspect the pump components on regular basis to avoid excessive temperature rise or loss of integrity of components, i.e. bearings, shaft.

The pump unit must be kept clean and in good working condition to ensure proper heat exchange with its surroundings.

2.8. Operating safety

ATTENTION!



The pump unit must not be operated in a manner incompatible with physical-chemical properties of the pumped medium, i.e. output, pressure, temperature, density, aggressiveness, abrasion, rotational speed as well as other parameters defined in pump technical data, offer details or other written arrangements with a manufacturer's representative.

2.9. Hazards resulting from temperature, pressure, flow and rotational speed

ATTENTION!

The user must ensure correct safety measures in the installation are provided so that temperature, pressure and rotational speed does not exceed applicable (minimum or maximum) limit values, permissible values or values that have been individually arranged.

Water hammer – In particular, the pump must be protected against results of its sudden stopping and possible water hammer impact. Thus it is recommended that, e.g. a check valve, tank with an air cushion, etc. should be used. Failure to fulfil this condition results in destroying the pump flow system and may result in housing damage.

System NPSH – In particular, the pump must be protected against excessive pressure drops in the suction pipeline below the $NPSH_r$ value plus a reserve equal to at least 10% of the $NPSH_r$ value, however, not less than 0.5 m. The required $NPSH_r$ value is individually specified for each operating point and provided in the operation characteristics. Failure to fulfil this condition results in cavitation in the pump, compromising the pump parameters and hydraulic system damage.

ATTENTION!	Thermal shock – In particular, ensure sudden temperature fluctuations are avoided as they may result in a thermal shock that may damage the pump. Reverse flow – In particular, it is necessary to protect the pump against reverse flow as it may result in increase of revolutions exceeding the permissible range and occurrence of tension on the electric motor terminals. Thus check valves must be used. Minimum flow – To avoid the risk of pump operation with a valve closed on the pumping side for a period longer that several seconds or hot water circulation, the system must be equipped with an installation ensuring minimum flow through the pump is maintained.
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3. DESCRIPTION NHVf TYPE PUMPS

3.1. Construction and operating principles

NHVf pumps are centrifugal single stage torque pumps. The pumps transfer mechanical energy of the rotating impeller to the medium present between its vanes. The centrifugal pumps cannot self-prime (transfer air through the pump and the suction line), and must be filled before every run.

NHVf type single stage centrifugal impeller pump with horizontal axis features a spiral channel with horizontal inlet (pump flows through the suction port) and a vertical discharge port (with axis corresponding to the pump axis). A centrifugal impeller is mounted inside the pump casing and fastened to the pump shaft with nuts.

The pump shaft is mounted on the bearings in the bearing mount using grease or oil lubricated rolling bearings. The pump is driven by an 50/60 Hz asynchronous motor at a rated speed of approx. 1000, 1450 and 2950 rpm (other models available depending on the intended use of pump unit i.e in marine versions , explosion-proof or with diesel motor).

The pump shaft is sealed with a cord seal or mechanical seal (single or double), lubricated and cooled with medium

The pump is mounted on the frame with lugs on the suction casing and the bearing casing.

Overall dimensions and rated parameters conform to z EN 733.

Suction and discharge inlets conform to ISO 7005-2/PN 16.

All impellers are balanced static and dynamic ISO 1940 class 6.3.

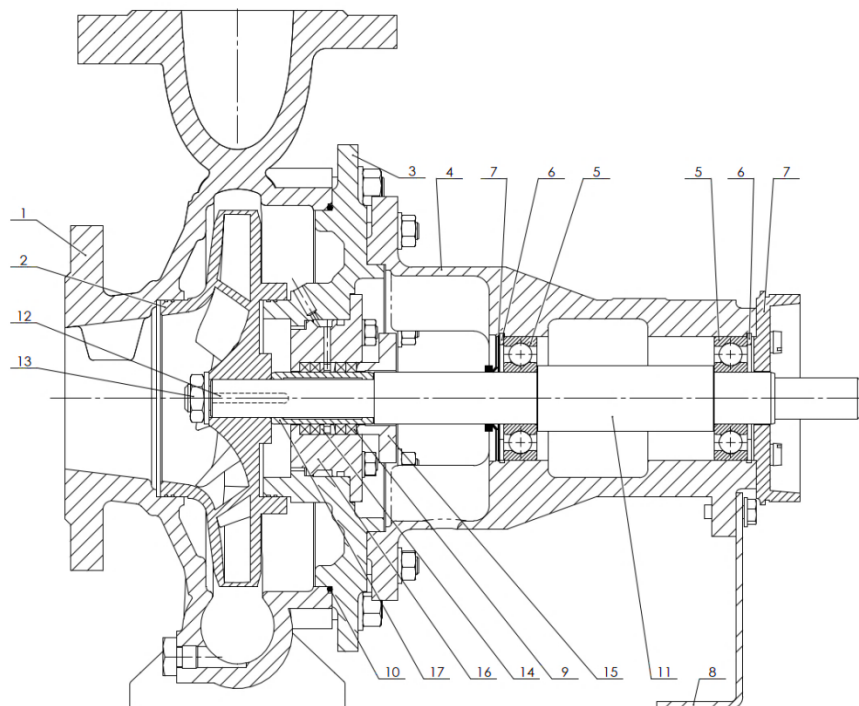


Fig . 3.1 Cross-section of NHVf with cord sealing and closed rolling bearings:

- 1 – spiral casing; 2 – impeller; 3 – sealing casing; 4 – bearing casing;
 5 – ball bearing; 6 – circlip; 7 – bearing cover; 8 – lugs;
 9 – cord; 10 – O-ring; 11 – shaft; 12 – key; 13 – nut;
 14 – spacer ring; 15 – gland; 16 – sealing sleeve; 17 – shaft protective sleeve;

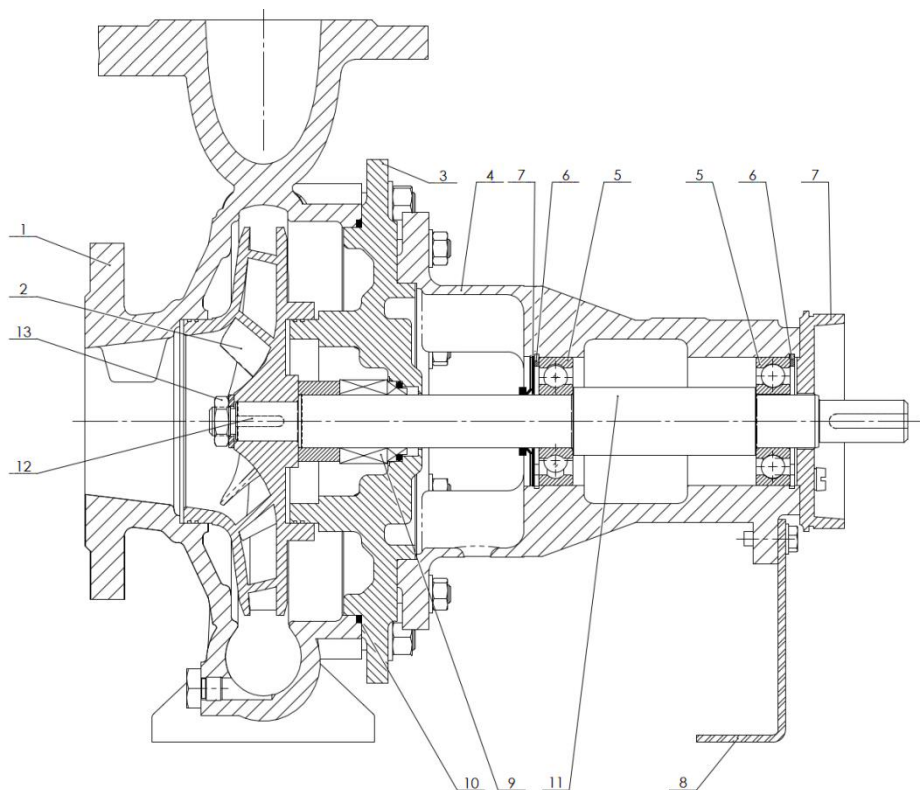


Fig. 3.2 Cross-section of NHVf series pump with mechanical sealing, and closed roller bearings:

- 1 – spiral casing; 2 – impeller; 3 – sealing casing; 4 – bearing casing;
 5 – ball bearing; 6 – circlip; 7 – bearing cover; 8 – lugs;
 9 – mechanical sealing; 10 – O-ring 11 – shaft; 12 – key; 13 – nut;

3.2. Applications

NHVf pumps are used for pumping clean or low solids content media (maximum solids content 20 mg/dm³) at up to 140°C, within the strength limits of materials used in production. Main applications:

- Industrial installations , pressure boosting , fire-extinguishing systems;
- Water systems, residential or drinking water,
- Irrigation and drainage systems,
- Fire extinguishing systems;
- Power engineering.

3.3. Technical data

Table 3.1 NHVf pump specifications

Pump capacity	to 1700 m ³ /h
Delivery head	to 100 m
Rotational speed	1000, 1500, 3000 rpm
Medium temperature	-15 to +140°C
Operating pressure	10 bar
Suction inlet diameter	DN 50 to 300 mm
Discharge inlet diameter	DN 32 to 250 mm
Max. ambient temperature	40°C

3.4. Marking of NHVf type pump

NHVf pump marking structure

Example: **NHVf-32-250/1.d.ee₁ee₂.h.iii.k**

NHVf-32-250	- Pump type
1	- Impeller turning (numeric marking for 3000 rpm, alphabetic marking for 1500 rpm version)
d	- Material execution
ee ₁ ee ₂	- Constructional execution
h	- Delivery completeness
iii	- Pump unit selection, manufacturer code
k	- Finishing (protective coats)

Tab. 3.2 NHVf pumps construction

Constructional execution	Constructional execution name	
e	1	for liquid temperature to +80°C
	2	for liquid temperature to +140°C
e ₁ e ₁	01	mechanical seal type Lidering 107
	02	mechanical seal type Anga A1
	03	mechanical seal type typu Anga V
	10	double mechanical seal type BED
	20	cord seal,
e ₂	0	standard sealed ball bearing
	1	reinforced bearing - closed bearings, holes for gauge plugs
	6	reinforced bearing - oil lubrication, holes for gauge plugs

Delivery Completeness

- 1 – pump without motor
- 2 – pump with coupling
- 3 – pump with coupling and base slab
- 5 – pump unit with base slab

Paint Coating

- 1 - standard
- 2 - special

Material execution

Table 3.3. Materials used for NHVf pump components

Part of the pump	Material execution								
	A	B	C	D	E	F	G	H	I
<i>Pumps casing</i>	Grey cast iron	Grey cast iron	Tin bronze	Acid-resistant steel	Ductile iron	Carbon steel	Chrome cast iron	Ductile iron	Carbon steel
<i>Seal casing</i>	Grey cast iron	Grey cast iron	Tin bronze	Acid-resistant steel	Ductile iron	Carbon steel	Chrome cast iron	Ductile iron	Carbon steel
<i>Impeller</i>	Grey cast iron	Tin bronze	Tin bronze	Acid-resistant steel	Tin bronze	Tin bronze	Chrome cast iron	Grey cast iron	Carbon steel
<i>Shaft</i>	Stainless steel	Stainless steel	Stainless steel	Acid-resistant steel	Stainless steel	Stainless steel	Stainless steel	Stainless steel	Stainless steel
<i>Bearing casing</i>	Grey cast iron	Grey cast iron	Grey cast iron	Grey cast iron	Grey cast iron	Grey cast iron	Grey cast iron	Grey cast iron	Grey cast iron

3.5. Shaft sealing

NHVf pump shaft is sealed with a gland seal with cord seal or front seal (mechanical). Front seal can appear in different variants.



Use of glandes with soft packing in explosion hazardous areas is prohibited.

3.6. Approximate noise level

Table 3.4 Approximate NHVf pump noise level

Motor power kW	Noise level, dB	
	1450 rpm	2900 rpm
< 0.55	60	64
0.75	60	66
1.1	62	66
1.5	63	68
2.2	64	69
3.4	65	70
5.5	66	71
7.5	67	73
11	69	74
15	70	76
18.5	72	77
22	73	78
30	74	79
37	75	81
45	75	82
55	76	82
75	77	84
90	78	85
110	79	85
132	80	86
160	80	86

3.7. Permissible loads and torques operating on the pump inlets

Forces and torques given in the table must not be exceeded.

Force and torque direction markings:

F_x – force directed along the X axis, i.e. along the pump drive shaft axis

F_y – force directed along the Y axis, i.e. perpendicularly to the pump axis and in parallel to the lug surface

F_z – force directed along the Z axis, i.e. perpendicularly to the pump axis and perpendicularly to the lug surface

M_x – torque around the X axis

M_y – torque around the Y axis

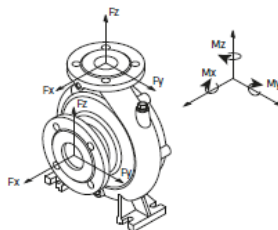
M_z – torque around the Z axis

Table 3.5 Permissible loads of NHVf pump flanges

Pump type	Discharge port									Suction port								
	DN	F _x [N]	F _y [N]	F _z [N]	Σ F	M _x [Nm]	M _y [Nm]	M _z [Nm]	Σ M	DN	F _x [N]	F _y [N]	F _z [N]	Σ F	M _x [Nm]	M _y [Nm]	M _z [Nm]	Σ M
32-125 32-160 32-200 32-250	32	315	295	365	565	385	260	295	550	50	575	525	465	905	495	345	395	720
40-125 40-160 40-200 40-250 40-315	40	385	350	435	680	455	315	365	660	65	735	645	595	1145	525	385	415	775
50-125 50-160 50-200 50-250 50-315	50	525	465	575	905	495	345	395	720									
65-125 65-160 65-200 65-250 65-315 65-400	65	645	595	735	1145	525	385	415	775	80	875	785	715	1375	555	395	455	820
80-160 80-200 80-250 80-315 80-400	80	785	715	875	1375	555	395	455	820	100	1175	1045	945	1835	615	435	505	905
100-160 100-200 100-250 100-315 100-400	100	1045	945	1175	1835	615	435	505	905	125	1380	1245	1115	2160	735	525	665	1120
125-200 125-250 125-315	125	1245	1115	1380	2160	735	525	665	1120	150	1745	1575	1400	2735	875	605	715	1280
150-200 150-250 150-315 150-400	150	1575	1400	1745	2735	875	605	715	1280	200	2345	2095	1890	3650	1135	795	925	1650
200-260 200-315 200-400 200-500	200	2095	1885	2345	3650	1135	805	925	1670	250	2925	2605	2365	4560	1555	1105	1275	2290

The above values apply to housing of gray cast iron (EN-GJL-250/GG25).

Larger values can be adopted for cast steels



3.3 Force directions

4. UNPACKING, TRANSPORT AND STORAGE

4.1. Unpacking

- After receiving the product, check if it complies with the waybill.
- Check if the packaging has not been visibly damaged during transport.
- Carefully remove the packing material. Check for any scratches or damage of the pump and other equipment (if any) that may result from improper transport.
- In case any damage is detected, report it immediately to the Customer Service Department.

4.2. Transport

4.2.1. General recommendations



- Keep strictly any applicable accident prevention regulations.
- During transport, always use protective gloves, helmets and boots with metal toe protectors.
- Wooden boxes, crates or cartons may be unloaded by means of forklifts or gantry cranes, depending on the packaging size, weight and design.

4.2.2. Lifting

- Before lifting and moving a pump, or a pump with a motor on a common base, always define:
 - total weight and position of the centre of gravity;
 - the maximum outside dimensions;
 - location of the lifting eyes.
- The lifting capacity of lifting equipment must comply with the pump or pumping unit weight.
- Both the pump and the pump unit must be lifted horizontally.
- It is absolutely forbidden to stay under suspended loads or in their vicinity.
- A load must never be suspended longer than it is absolutely necessary.
- Acceleration and braking during lifting operations must be safe for persons.

Lift the pump or the pump unit as shown in Fig. 4.1a and 4.1b to avoid obstacles (do not use motor lugs to lift the entire pump unit).

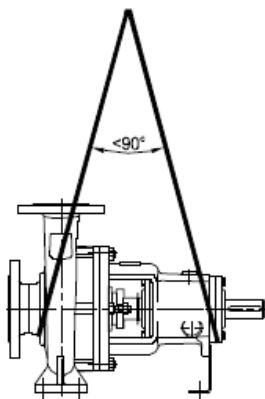


Fig.4.1a Pump with free shaft end

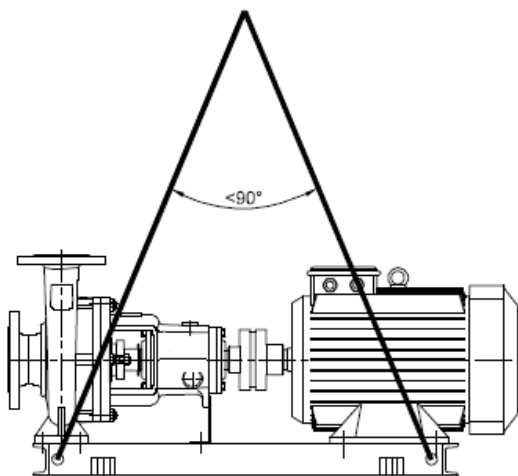


Fig. 4.1b. Pump and motor on a common frame

4.3. Storage

- If a pump is to be mounted and commissioned right after delivery, it must be stored in a clean, dry room, protected against frost and ensuring moderate temperature fluctuations.
- If a pump is equipped with bearings which require lubrication, apply an extra layer of grease in order to avoid moisture penetration into the area around the shaft.
- The pump must be adequately secured against moisture, dust, dirt and foreign bodies.
- The pump shaft must be periodically turned (for example once a week), to avoid bearing surface pitting and pump seizure.

5. ON-SITE INSTALLATION

ATTENTION!	The device must be installed in accordance with the regulations. The pump may be installed, levelled and adjusted only by experienced personnel. Faulty installation or incorrect foundations may result in operation issues not covered by the warranty.
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5.1. Pump with shaft without cover

- If a pump is delivered with a uncovered shaft end, it is necessary to prepare a correct plate on which the pumping unit with a motor is to be installed. The base plate must be designed and constructed in a manner ensuring necessary stiffness and vibration prevention.
- If a pump is delivered without an electric motor and coupling, prior to installing the unit, select a correct motor and coupling.

While selecting a motor, the following factors must be taken into account:

- maximum power received by the pump within the whole operating range;
- pump rotational speed;
- available power supply (mains frequency, voltage, etc.);
- motor type (explosion-proof version etc.);
- motor installation location (installed on lugs).

While selecting a coupling, take into account motor power, operating speed.

5.2. Preparation for installation

Before the pump is installed:

- Carefully clean suction and discharge flange.
- Remove the protective layer from the pump shaft.
- If a pump has been stores for some time, remove grease from bearings or oil from the bearing housings (if the pump is lubricated with oil). Then clean bearings with a correct cleaning solution and re-lubricate.

NOTE: This is not required when the bearings are lubricated once for their lifetime.

5.3. Installation site

ATTENTION!	A pump must be installed in a location protected against frost and dust. The surroundings must be well-ventilated, without explosive atmosphere (with the exception of pump units for use in hazardous areas). A pump must be installed is a manner ensuring access for maintenance purposes and the room must be ventilated as appropriate. The clearance above the pump must make it possible to lift it. The suction pipe must be as short as possible. No redundant fittings that may lower the system net positive suction head should be fixed on it.
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5.3.1. Foundation

ATTENTION!	Foundations must be prepared and pump unit must be installed with utmost care. Incorrect installation results in premature wear of pump components and its malfunction. Foundations should be heavy enough to reduce vibrations and stiff enough to avoid warping and mismatch. Concrete must fully set before the device is installed. The foundation surface must be levelled correctly and as smooth as possible.
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5.3.2. Installation

- Locate the pumping unit on the foundation and add/remove shims under the base plate to adjust the level of the outlet orifice, using a spirit level according to Fig. 5.1. The unit must be ideally levelled.
- Lightly tighten the anchor bolts.
- Check the coupling adjustment.

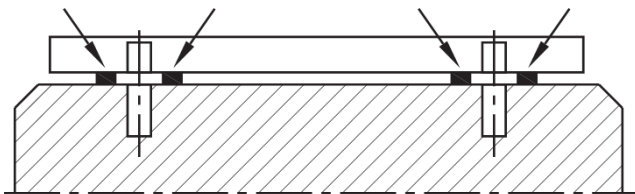


Fig.5.1. Foundation, base and installation of shims

- Fill the base plate with concrete. Remove all air bubbles. The plate must be tightly integrated with the concrete foundation.
- Wait until concrete sets (at least 3 days).
- Tighten the foundation (anchor) bolts. Check coupling alignment.

5.4. Alignment of the coupling

ATTENTION!	Proper coupling regulation is the most important factor ensuring problem-free operation of the pumping unit, during the whole operation period. Incorrect regulation is the main reason for vibrations, noisy operation as well as overheating and overloading of bearings. Thus, the coupling must be correctly aligned and its alignment must be frequently inspected
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- A flexible coupling does not correct the alignment. A flexible coupling is able only to compensate for slight misalignment. Maximum misalignment must not exceed values in Table 5.1.

ATTENTION!

The "coupling alignment" term means setting the motor and pump rotation axes in one straight line. NHVf series pumps are delivered with/after pre-alignment (if delivered with a drive and foundation plate). The plate may be deformed during transport and handling. Re-align during installation.

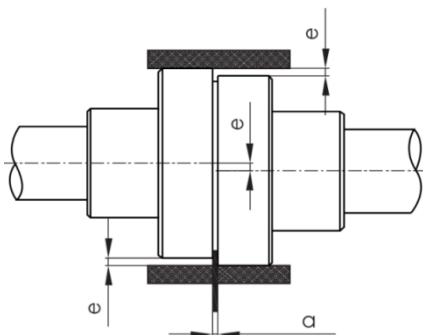


Fig.5.2. Flexible coupling alignment

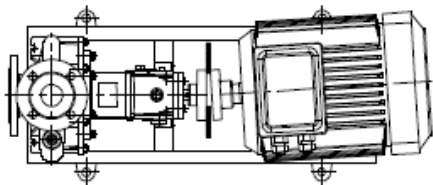


Fig.5.3a: Angle deviation in the horizontal plane and seating deviation

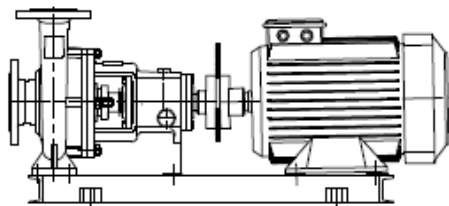


Fig.5.3.b: Angle deviation in the vertical plane and seating deviation

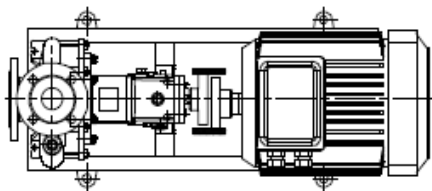


Fig.5.3c: Horizontal parallelism deviation and set-back deviation

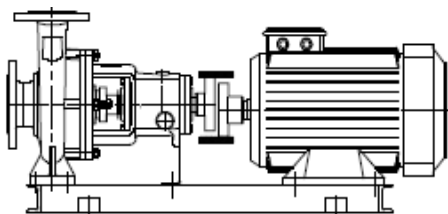
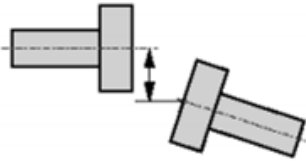
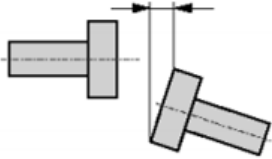


Fig.5.3d: Vertical parallelism deviation and set-back deviation

- To re-align the coupling it is recommended to use laser devices or straight edge
- There may be two different types of coupling alignment errors:
 - a) angle deviation;
 - b) parallelism deviation
- While inspecting the angle error, the gap between two halves of coupling discs must be equal, both vertically and horizontally.
- For checking the straight edge to a half of the coupling plate, keep it parallel to the axis and its position in relation to the other half to check the parallelism deviation. The straight edge must touch both coupling disc halves precisely in all points, in vertical and horizontal plane (Fig.5.2).
- Misalignment errors may occur both in the horizontal and vertical plane. In the vertical plane, misalignment is corrected by inserting adjustment washers under the motor or pump foot. However, in the horizontal plane, corrections are made by shifting a pump and/or motor horizontally by means of the installation openings. Drawings 5.3a, 5.3b, 5.3c and 5.3d show the step-by-step procedure of coupling adjustment.

Table 5.1 Approved values for pumping engine misalignment

Deviation	RPM	Acceptable value	Perfect value
<i>Parallel shift (e)</i> 	1000	0,14	0,09
	1500	0,09	0,06
	3000	0,06	0,03
<i>Angle shift (a) (gap width in reference to 100 mm of coupling diameter)</i> 	1000	0,11	0,08
	1500	0,07	0,05
	3000	0,04	0,03

ATTENTION!	After making all changes, re-check the alignment as each shift in one direction may cause misalignment in another direction.
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5.5. Connecting a pipeline

5.5.1. General

ATTENTION!	It is not allowed in any case to use a pump as an anchoring point or support for a pipeline.
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- Pipes should be supported very close to the pump (Fig. 5.4a and 5.4b). Make sure that pipeline loads and tensions are not transferred to the pump. Thus, after pipeline installation, loosen the suction and discharge flange bolts to prevent stress transfer from the line to the pump.
- Nominal measures of flanges on the discharge and suction side are not reference points for selecting correct dimensions of the suction and discharge pipelines. Nominal pipe diameters should be equal or larger than the pump housing dimensions. Do not use pipes and accessories with diameter lower than the pump port diameter. Select the foot valves, strainers, filters and check valves with higher flow areas. Generally, linear flow rate shall not exceed 2 m/s at the suction line and 3 m/s at the discharge line. Higher flow rates will increase pressure drops, which may result in cavitation at the suction line and friction losses at the discharge line.
- Pipes must be connected by means of flanges with seals of correct size and made of appropriate materials. A flange seal must be located centrally between flange bolts, such a way that the opening is not obstructed.
- Take into account thermal expansion and excessive pipeline vibrations so that their effects are not transferred on the pump.
- The pipe must be cleaned from welding residues, sand, rust and possible hurds, which may be deposited therein. Plug the pump inlets in order to prevent dirt penetration After the system is assembled, all pipeline parts must be dismantled, thoroughly cleaned, painted and reassembled. If a sieve is used on the suction side, it should be cleaned after several days of operation.

ATTENTION!	It is recommended to use supports and expansion joints on the suction line and discharge lines, in order to reduce stresses on the system.
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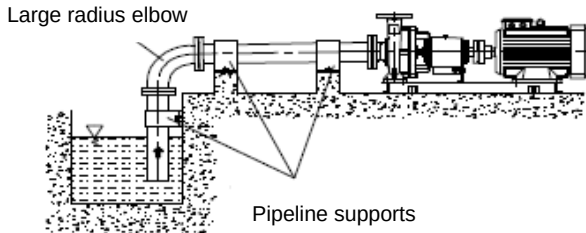


Fig.5.4a Pump in suction configuration

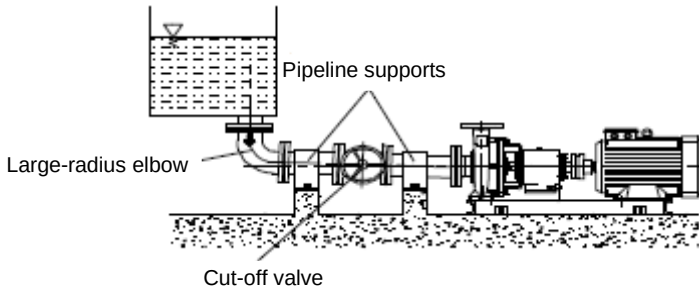


Fig.5.4b Pump in inflow configuration

5.5.2. Suction and discharge pipeline

- The suction pipeline must be absolutely leak tight and located in a manner preventing air pocket generation. The suction pipeline must be located with a slight head towards the pump suction installation with inflow (Fig.5.5b) or should be slightly lifted towards the pump suction installation with suction (Fig.5.5a).
- In order to maintain possibly low losses caused by friction, it is necessary to avoid sharp bends and sudden changes in flow directions or cross-section sizes, and providing the shortest suction pipe as possible. If it is necessary to change the cross-section of almost horizontal placed pipeline, provide an eccentric reducer with a horizontal upper surface.
- Inflow line upstream of the pump inlet should guarantee uniform velocity field at the pump inlet. In order to that Install a straight run of a length equal to at least five times the line diameter. The line length depends on the obstacle type upstream of the straight run. A longer straight run may be required for elbows with a low bend radius, tees and partially open valves.
- The inflow pipeline, providing the anti-cavitation allowance, should be fitted with a cut-off valve with a stem located horizontally. This valve must always be opened during pump operation and must not be used for flow adjustment.

- A control valve should be mounted on the discharge pipeline as close to the pump as possible.
- If the pump achieves at least 10 m delivery head or the discharge line is long, install a check valve between the pump and the control valve.



In explosion hazardous areas in particular, it is necessary to guarantee a continuous medium flow to the pump and prevent foreign objects from entering the pump. For this purpose, use strainers at the suction line, which require periodical cleaning or replacement and must guarantee low flow resistance.



To prevent the explosive atmosphere from forming when transferring flammable media, make sure the pump and lines are filled with the medium at all times.

If the pump and the lines cannot be filled with medium at all times, use suitable monitoring (e.g. inlet pressure measurement, medium level sensor, dry run sensor) to prevent dry run and temperature rise above the permissible limits.

5.5.3. Additional pipe connections and accessories

- Depending on the application, it is possible to install additional connections (for cooling, closing, liquid draining) and additional operating condition measuring devices.
- Positive and negative pressure gauges must be correctly fixed and connected to measurement points of max. diameter approx. 8 mm, located on pump flanges or pipes (placed as close as possible to flanges) and set up in a spiral configuration to reduce pressure fluctuations. For safety reasons, a cut-off and venting valve should be installed upstream of the gauges (Fig. 5.5).

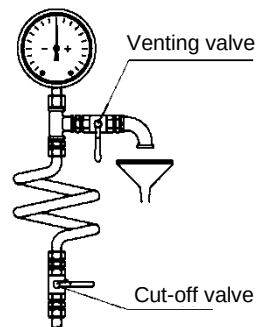
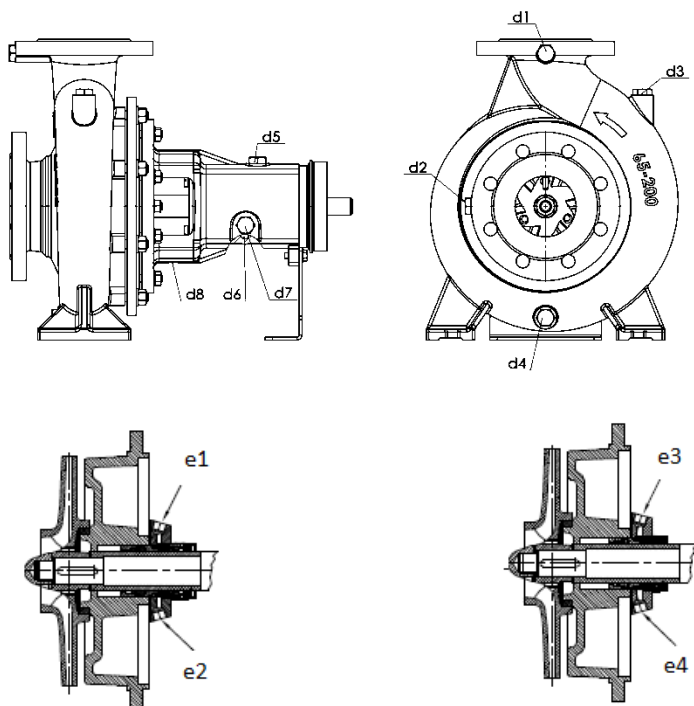


Fig.5.5. Pressure gauge installation

- Each pump is fitted with drain connectors/plugs at its casing and connectors at the bearing mounting to reduce the gland leakage (Fig. 5.6). If it's necessary to drain the pump or remove leakage, it's possible to connect pipe connectors to appropriate tank using pipes. The draining tubes must be equipped with a cut-off valve and must be adjusted to the pump maximum operating pressure.
- Cooling pipes, sealing and rinsing the packing must only be connected to connectors provided for this purpose and located on the pump.



d1: Static pressure (outlet)

d2: Static pressure (inlet)

d3: Air release

d4: Drain plug

d5: Oil filler plug

d6: Oil drain plug

d7: Oil filling device

d8: Packing leakage

e1: External buffer liquid - inlet, dual mechanical seal

e2: External buffer liquid - outlet, dual mechanical seal

e3: External buffer liquid - inlet, single mechanical seal

e4: External buffer liquid - outlet, single mechanical seal

Fig.5.6. Locations of connectors/plugs

5.5.4. Minimum flow

If there is possibility that the pump may operate with zero flow (closed discharge valve) or with almost closed valve, i.e. practically without flow, a minimum flow valve (or a closed bypass valve) should be installed on the discharge side, right after the pump, but before a gate valve/flow regulating valve. If no valve is installed or the valve is closed for a long time, the pump may be damaged, since all the motor power is transformed into thermal energy absorbed by the medium.

5.5.5. Electric connections



Electric motors must conform to (EN 60034-1).

Enclosures of electric motors and control systems for the pumping unit must ensure at least the IP22 protection degree. Nevertheless, while defining the electric motor and control system housing protection degree, take into account the operating and environmental conditions.



In potentially explosive atmospheres use only motors which are allowed to work in hazardous areas .



In order to prevent accumulation of electrostatic discharges, provide the pumping unit with an earthing system in the form of a electro-conductive wire of resistance below 1 Ω connected to the unit's frame.

- Prior to commencing any work disconnect the power supply.
- The supply cable must be routed in such manner that it does not touch any pipeline, pump and motor enclosure.
- Check if supply parameters (frequency, voltage and phases) given on the data plate comply with the parameters of the mains.
- An electric motor must be protected against overload by means of isolating switches and/or fuses. Isolating switches and fuses must be selected in line with current drawn at full load (see the motor data plate). Protect the motor against voltage fluctuations and unbalance.
- It is recommended to use PTC sensors in the motor, however, it is an optional measure depending on customer's requirements. If PTC sensors are used, this system must be connected via suitable connectors in a junction box. A PTC sensor must be connected to a mechanism ensuring isolation in the event of overheating.



Listed below are typical diagrams used to connect power supply and sensors located in motors!

Individual connections of delivered motors must be made in compliance with information provided on the motor and documentation submitted by a given motor manufacturer or available on their website!

5.5.6. Standard connection of motor sensors

The type of sensors located in a motor is defined on the data plate. When motor is equipped with temperature sensors, they must be connected to the control system. If temperature sensors are not connected to the control system, the motor warranty becomes void.

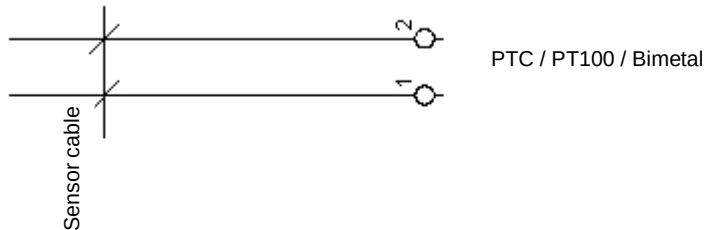


Fig.5.7.Single sensor connection diagram

PTC sensor – Usually, these are three sensors connected in series, i.e. one per each power supply phase. PTC sensors are selected by the manufacturer in relation to temperature compliant with the motor insulation class. There are two cables which must be connected to a converter mounted in the control cabinet whose contact is used in the pump motor control line. PTC sensor power supply - max. 7.5 V DC.

PT-100 sensors – 1, 3 or 6 such sensors may be mounted in motor windings; additionally, they may be installed in pump bearings. As regards motors equipped with 6 PT-100 sensors in a winding, it is necessary to connect three of them, separately for each phase. Each PT-100 sensor must be connected to an individual converter which converts resistance values into temperature values. For example, the temperature of motor shutdown for class F insulation is 135°C for sensors in a winding (20°C below working temperature due to the possibility of spot overheating).

The limit temperature for bearing operation is typically 90°C. However, the manufacturer may allow for a higher operating temperature.

Motor temperature classes:

Class A - operating temperature up to 105°C; setting 85°C

Class E - operating temperature up to 120°C; setting 100°C

Class B - operating temperature up to 130°C; setting 110°C

Class F - operating temperature up to 155°C; setting 135°C

Class H - operating temperature up to 180°C; setting 160°C

Bimetal sensor – It emits a binary signal:

1. short-circuit – permission to operate
2. break – motor overheated; must be stopped

5.5.7. Connecting motor power supply

- Before connecting electric wiring, manually turn the pump shaft to check if it rotates freely.
- Electric wiring must be connected in line with local regulations. Hazardous voltage on the motor housing must be prevented.
- The connection diagram is located in the junction box or motor operating manual.

Typical connections of electric motors

a) Factory-paired motor - 3-phase voltage

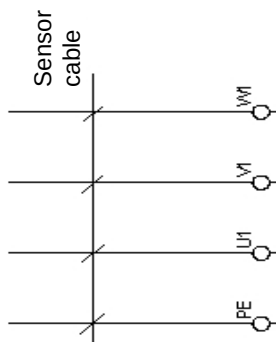


Fig.5.8. Connection diagram for a motor factory-paired for 3-phase voltage

- b) Star-connected motor (designation on the data plate - motor adapted for star connection or nominal voltage for 400 V AC mains is 230/400 V, which means that the interphase voltage for a delta connection is 230 V AC and 400 V AC for a star connection).

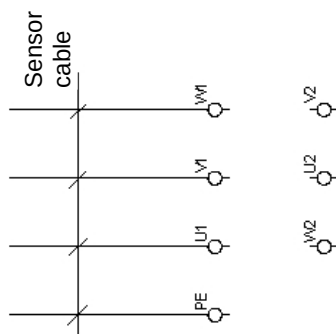


Fig.5.9. Connection diagram for a star-connected motor

- c) Delta-connected motor (marking on the data plate - motor adapted for delta connection or nominal voltage for 400 V AC mains is 400/690 V, which means that the interphase voltage for a delta connection is 400 V AC and 690 V AC for a star connection)

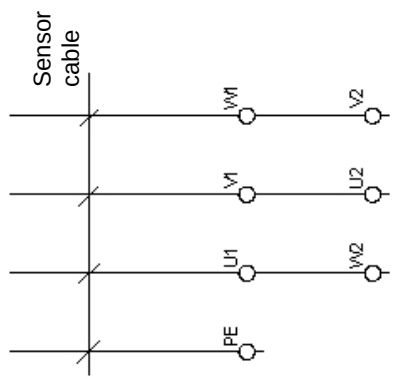


Fig.5.10. Connection diagram for a delta-connected motor

Motors wound to operate in a delta connection may be supplied from star/delta switches. Then, factory-installed jumpers are not used - see above. In such cases, terminals W2, U2 and V2 are supplied with another cable, and the control cabinet must contain contactors which switch the connections type. They must be connected according to applicable best practices.

Table 5.2 Recommended motor connection type according to rated motor power

Connection type	Rated motor power PN ≤ 4 kW	Rated motor power PN >4 kW
	Supply voltage 3~400 V	Supply voltage 3~400 V
Direct	Star connection (Y)	Delta connection (Δ)
Y-Δ switch	Impossible	Possible

- A connection diagram depends on the nominal motor power, power supply and connections type.



As regards 3-phase induction motors with a $Y-\Delta$ connection, the connection must be made in such a manner that the time of switching between a star and delta is short. **The start-up time in the star connection must be determined experimentally so that a motor with a star connection achieved revolutions similar to nominal ones. Switching time that is too long may cause pump damage. See the table below for typical operation time limits for a star connection. The time given is only indicative and must be checked during unit start-up.**

Table 5.3. Typical star-delta switching times

Motor power	Y stabilisation time
≤ 30 kW	< 3 seconds
> 30 kW	> 5 seconds

5.6. Motor operation with a frequency converter.



Possible motor operation with a frequency converter must be agreed upon while ordering the unit. In motors (particularly as regards power above 75 kW) operating with a frequency converter, insulated bearings must be used.

The permissible unit rotation adjustment range by adjusting the frequency of the voltage supplying a motor depends on individual features of a pump and is always within 30 Hz - 60 Hz. When the frequency value is increased above the nominal value, the nominal value of motor current must not be exceeded. When it is expected that operation is to be conducted with frequency higher than the nominal value, a motor of a higher nominal value must be selected. Moreover, it must be confirmed with the pump manufacturer that a pump may operate with the required frequency.

5.7. Final inspection

- After completing all activities described above, re-check the coupling alignment in line with section 5.4. Correct it if necessary. Turn the impeller by hand to check if it rotates freely. Install all safety measures, start the unit and wait until it warms up and reaches typical operating parameters. Deactivate the unit and make all necessary final alignment adjustments. **The final fitting must be conducted in the operating temperature.**



Install guards. Do not commission the pump until the guards are mounted.
This is necessary to ensure operation safety.

6. START-UP, OPERATION, STOPPING



The pump installation may be activated only by personnel trained in installation and operation of the pump and system. The personnel must also be familiar with applicable local regulations.

6.1. Preparation for start-up

- Standard NHVf pumps are equipped with sealed bearing which are filled with a grease for a whole operating time. In case of pumps equipped with open bearings check for fouling during transport or installation before first run.
- Check the level of the pumped medium in the pump. NHVf pumps and suction line must be primed with medium.
- Check if elements rotate freely and evenly.
- Check if the pump, motor, coupling guard as well as other additional and protective devices have been installed correctly.
- Check if the suction line valve is open, the discharge line valve must be opened approx. 25% (pumps up to 30 kW can be run with the discharge line valve temporarily closed).
- Check if the electrical connections are made correctly.
- Turn the pump on/off for a short time to check the rotation direction. Motor should rotate clockwise as viewed from the drive side.

After commissioning:

- Define the pump operation set point in line with the data plate or data provided in the order form

ATTENTION! 	Operation with a closed valve on the suction or discharge side is forbidden as it may result in excessive increase in temperature and vibrations resulting in pump damage.
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- While commissioning the pump with an empty discharge pipeline, until the pipeline is fully filled, partly close the discharge valve so that the pump operates within the permissible operating range.

ATTENTION!	Operation with an empty discharge pipeline may result in excessive increase in output, which causes motor overloading, cavitation and increased vibrations.
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- Adjust the leakage from gland seal (if there is one). Correct leakage should amount to ~60 drops per minute, re-check after hour of work operation. To adjust the leakage tighten evenly the screws fixing the gland seal.
- If there's no possibility for further tightening of gland seal stop the pump and replace cord for new one. Cut cord at an angle 45° and mount in gland seal alternately.

	Access to gland pacing requires removing the cover. Special care should be taken during regulations, long hair should be tied and hidden, clothes should not be loose-fitting , its creating the risk of incidental drawing them in on a shaft. After regulations mount the cover.
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ATTENTION!	Incorrect adjustment of gland seal or lack of it causes damage to sealing cord and shaft protecting sleeves, increase of power input or excessive leakage.
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- Evaluate the mechanical seal performance. While working on the mechanical sealing, act in line with individual manufacturer's recommendations. Unless it is recommended otherwise, during commissioning, check if no liquid penetrates the packing. If it is so, the packing can be damaged. During normal operation, the leakage from the mechanical seal is invisible and does not require any adjustment.



In case the required set-point or parameters recommended by the manufacturer are not achieved, there are abnormal noises or excessive vibrations as well as leakages from locations other than the gland packing, immediately shut down the pump and look for reasons for such behaviour.

6.2. Restarting

Before the pump is restarted, follow all procedures provided for commissioning, except for: inspection of bearings, correct installation and rotation direction.



Exercise caution as the pump may still be hot after previous operation.

6.3. Pump operation limits



During operation, always observe the limits concerning permissible temperature, pressure, output and speed described on the data plate or datasheet.

Operating a pump with parameters other than the nominal ones results in its premature wear. In particular:

- Do not exceed the motor nominal power.
- Do not exceed for prolonged periods the recommended operating parameters (in particular, the output values). Unless specified otherwise, operation must be within the following range: $0.7 \div 1.2$ QBEP (optimum output), provided that the required system NPSH is ensured.
- Do not use the pump in systems of pressure higher than permissible.
- Do not continue operation of the pump works unevenly or with excessive vibrations.
- Do not use the pump to pump media of abrasive properties exceeding the permissible values.

ATTENTION!	Abrasive pollutants cause faster wear of sealing elements of the pump. Frequency of inspections must be increased in comparison with operating media without abrasive content.
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The permissible number of pumping unit activations is determined by the drive capabilities. Observe the instructions provided in the drive motor operating manual. In case such data is unavailable, use the chart shown below.

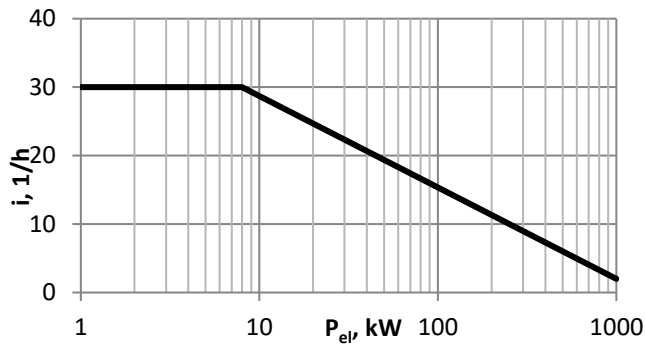


Fig. 6.1. Permissible number of electric motor activations per hour

6.4. Operation monitoring

ATTENTION!	Bearing temperature may exceed ambient temperature by 50°C. Do not exceed ambient temperature by 80°C.
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In areas with danger of explosions constant monitoring of the bearing temperature and the bearing mount vibrations is recommended.

It is recommended to periodically monitor a pumping unit to mitigate the malfunction risk, in particular:

- Once a week, conduct a visual inspection for pump leaktightness.
- At least once a week control leakage from cord gland.

ATTENTION!	Pumps subject to excessive corrosion or abrasive wear should be inspected more often than in standard operating conditions. The first inspection for wear should be conducted after 6 months. Next inspections depend on individual operating conditions.
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6.5. Stopping the pump

- When a pump is stopped in an installation without a check valve, close the valve on discharge line and immediately deactivate the pumping unit.
- Check if any atypical noises are heard while stopping the motor.
- After complete stop, close the suction and discharge valve.
- As regards pumps subject to low temperatures and possibility of medium freezing inside the pump, drain the liquid by removing all plugs in the lower section of the pump. Additionally, drain the liquid from pipelines.

6.6. Lubrication

- NHVf pumps are grease lubricated; oil lubricated versions are also available.
- NHVf pumps are usually equipped with sealed bearings lubricated by permanent grease filling.
- Oil change interval
 - at 2900 rpm: 1500 hours of operation
 - at 1450 rpm: 3000 hours of operation
- Recommended oil HIPOL 15F

Table 6.1 Recommended oil amount – reinforced bearing, oil lubricated

<i>Pump size</i>	<i>Shaft diameter, m</i>	<i>Bearing type</i>	<i>Oil volume, dm³</i>
<i>A</i>	24	6307	0,45
<i>B</i>	32	6309	1,4
<i>C</i>	42	6311	1,75
<i>D</i>	48	6313	3,5

- Check oil level and top up as necessary. Once a year drain the oil chamber, clean and fill with new oil.
- Do not reuse any disassembled bearings.

7. SERVICE AND MAINTENANCE



Any servicing must be carried out by a trained and experience personnel familiar with the content of this operation and maintenance manual.

Make sure the pump unit's power supply is disconnected and protected against accidental activation before commencing any works on the pump unit.

Wait for the pump casing temperature to reach ambient temperature for high or low temperature medium.

Close the discharge and inflow line valve and equalize the pump and ambient pressure and drain the medium before disassembly.



For corrosive or hazardous media, rinse the pump with inert fluid or decontaminate and dispose of the medium and washings in accordance with relevant regulations.

Drain the bearing casing in pumps with oil lubricated bearings.



In potentially explosive atmospheres it is prohibited to use tool which usage may lead to sparks and fire, use non-sparking tools only.

ATTENTION!

The manufacturer recommends use of genuine spare parts purchased from an authorized service centres or directly from the manufacturer. Compare replaced and new parts and make sure they are identical and made of the same materials.



The bearing retainer in pumps for use in potentially explosive areas, should be inspected daily for excessive heat and noise, as well as grease or oil condition. Change grease or oil immediately after observing any wear to the bearings.

During work observe safety regulations and secure the pump and its components against accidental falls.

During the disassembly of the pump as a result of seal damage, it is recommended to check all other components, e.g. bearings and replace if necessary.

The components must be cleaned, and all mating surfaces must be degreased and smooth.

The required bolt torque values must be respected.

Installation and mating surfaces may be smeared with inert substance e.g. soap to facilitate installation.

7.1. Gland with soft sealing

Gland with soft sealing requires permanent supervision which should be expressed by checking the amount of leaking liquid. Correct leakage through gland seal should amount to 60-90 per minute.

ATTENTION!

Liquid from gland seal must not flow and water vapour should not get outside, as this could result in burns.



Slight leakage may cause overheating of gland seal, damage of cord and damage of shaft protecting sleeves. Major leakage may cause excessive increase of vibrations and pump bearing can be flooded.

If further gland seal regulation is not possible, packing should be replaced with new one. Work only when pump is stopped and when there's no working medium.



When pumping aggressive liquids rinse pump before dismantling the sealing.

Cut the ends of cord at an angle 45° and lay it in gland seal turned 90°, The cord length should be adjusted so that there's no gaps between its ends. Before pump commissioning with new seal packing lightly tighten using gland seal without causing shaft locking. After start-up adjust leakage to amount ~60 drops per minute and install the seal shield.

7.2. Gland with mechanical sealing

Mechanical seal of the pump fully reduces leakages. In case they are noticed, it means that it has been damaged.

As a rule, gland with mechanical seal are maintenance-free. Maintenance of specialist packing equipped with support systems must be conducted in line with packing manufacturer's recommendations and the manual provided.

See section 8 for packing replacement procedures.

Mechanical seals can be used only when the pump is primed with liquid.

ATTENTION!	Mechanical seal must not be dry-operated as it results in their immediate damage.
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When packing is replaced, drain liquid from the pump and, if the liquid is aggressive, rinse with warm water. All activities must be conducted during standstill, after the unit is secured against unintentional activation.

During new packing installation, thoroughly clean the chamber. In order to facilitate the process on inserting the packing on the shaft and housing, they may be covered with soapy water. Do not touch the frictional surfaces of sliding rings. In case any contamination is noticed, remove it with acetone. Prior to installation, check the sealing rings for visible scratches. In case any irregularities are detected, replace the packing with new one that is free of flaws.

Depending on the application, NHVf pumps are also available with different seal types.

7.3. Bearings

As a standard, NHVf series pumps come with rolling bearings lubricated with solid grease. Bearing versions lubricated with oil are also available. The minimum design life of bearings is 20,000 h. After this time, bearings must be replaced.

In hot water pumps, bearings made of bearing steel for operation in high temperatures are used.

ATTENTION!	Bearing temperature may exceed ambient temperature by 50°C. Do not exceed ambient temperature by 80°C
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The bearing housing in pumps for use in potentially explosive areas, should be inspected daily for excessive heat and noise, as well as grease or oil condition. Change grease or oil immediately after observing any wear to the bearings.

For oil lubricated pumps, check oil level in the bearings and top up as necessary. At least once a year, drain the spent oil and refill with new oil, up to the indicated level.

7.4. Coupling

Regularly (every 1000 hours) check the gap between coupling elements and unit alignment. In case there is misalignment, re-align the unit (Table 5.1).

ATTENTION!



Check coupling components (inserts, rubber etc.) every 1000 hours, replace worn components and check unit alignment.

7.5. Maintaining the pump cleanliness



Presence of large amounts of dust and dirt on external surfaces of the pump results in its incorrect cooling and may cause excessive temperature increase.

ATTENTION!

Always keep the pump clean. Do not use pressurised water to clean the pump as it may penetrate the rolling bearing housings

8. DISASSEMBLY, REPAIRS AND REASSEMBLY

8.1. General



All service and maintenance activities concerning a pumping unit must be conducted only by trained and experienced persons who are familiar with the content of this operating manual!

Before commencing work on the pumping set, ensure it is disconnected from power supply, locked out and tagged out.

As regards pumps operating low- or high-temperature liquids, wait until the housing reaches the ambient temperature level.

Before dismantling a pump, close the valve on the discharge and suction inflow side, equalise the pump pressure with the surroundings and drain the pumped medium.



As regards pumps operating aggressive or hazardous liquids, their interior must be rinsed with a neutral liquid or disinfected. The drained liquid and washings must be handled inline with applicable regulations.

As regards pumps with oil-lubricated bearings, drain oil from bearing housings.



In potentially explosive atmospheres it is prohibited to use tool which usage may lead to sparks and fire, use non-sparking tools only.

ATTENTION!

Use only parts purchased in recognised service outlets or directly from the manufacturer. Before assembly, new parts must be compared with the dismantled parts to make sure they are equivalent and made of the same material.

During work, observe all safety instructions and secure the pump and its components against accidental dropping.

Do not use excessive force. Elements subjected to force, e.g. hit with a hammer, must be secured with a pad made of plastic, hardwood or copper.

It is recommended that, during damaged (e.g. damaged packing) pump disassembly, inspection of the damaged element as well as all other details, e.g. bearings, should be inspected and replaced if worn.

The installed elements must be clean and the contact surfaces must be additionally degreased and smoothed as required.

Observe the required bolt tightening torques.

In order to facilitate assembly, the installation and contact surfaces may be covered with a neutral agent, e.g. soap.

8.2. Tools and equipment

In order to conduct service activities, usually standard workshop equipment is sufficient. However, in order to facilitate the required work, the following elements are recommended:

- set of box spanners,
- set of socket wrenches,
- ratchet spanner,
- screwdriver,
- light hammer
- lever,
- circlip pliers
- torque wrench,
- bearing puller,
- induction heater for bearings

In order to degrease the contact surfaces, use appropriate agents, e.g. acetone.

In order to facilitate installation of elastomers, use soapy water or liquid soap.

If there are problems with loosening threaded or driven connections, use suitable penetrators. If penetrators are used, replace elastomers as they may become damaged.

8.3. Bolt tightening torques.

Table 8.1 Recommended bolt tightening torques

<i>Thread diameter</i>	<i>Bolt tightening torque, Nm</i>	
	<i>Strength class</i>	
	<i>8.8</i>	<i>10.9</i>
<i>M4</i>	3.0	4.4
<i>M5</i>	5.9	8.7
<i>M6</i>	10	15
<i>M8</i>	25	36
<i>M10</i>	49	72
<i>M12</i>	85	125
<i>M14</i>	135	200
<i>M16</i>	210	310
<i>M18</i>	300	430
<i>M20</i>	425	610
<i>M22</i>	580	820
<i>M24</i>	730	1050
<i>M27</i>	1100	1550
<i>M30</i>	1450	2100
<i>M33</i>	1970	2770
<i>M36</i>	2530	3560

8.4. Disassembling NHVf pump unit

- Disconnect motor power supply.
- Close suction and discharge line valves.
- Disconnect any accessory cables.
- Drain the system - remove drain plugs at the bottom of the pump casing.
- Remove the screws and the coupling cover.
- Remove the bolts securing the motor to the base plate.
- Use a lifting eye bolt to lift the motor from the base plate.
- Remove the coupling insert.
- Remove the bolts securing the pump to the base plate.
- Use straps attached under the pump casing to lift the pump unit from the base plate.
- Loosen the coupling hub set screws.
- Remove the coupling hub from the pump shaft with a puller.
- Remove the key.
- Remove the coupling hub from the motor shaft with a puller.
- Remove the key.

8.5. Disassembling NHVf pump with cord sealing

Refer to Fig. 12.1 for disassembly of the pump with cord seal.

After separating the motor from the pump and transporting to workstation start the dismantle.

- Remove nuts (37) fixing spiral casing (219) and seal casing (208).
- Remove spiral casing (219).
- Remove nut (39) securing the impeller (1).
- Slide the impeller (1) from the shaft (225) and remove the key (44).
- Remove nuts (212) tightening the gland (217) to seal casing (206).
- Remove nuts (38) connecting bearing casing (221) with sealing casing (208).
- Remove loose sealing casing (208).
- Remove the cord sealing (205) along with spacer ring (207) and gland (217).
- Remove bearing covers (222), (223) and retaining rings (228) using the pliers.
- Remove shaft (225) with bearing (227) from bearing casing (221).
- Remove bearings (227) from shaft(225) using puller

Assemble in the reverse order. Carry out assemble of bearings on the shaft through heating the bearing to temperature from 80°C to 100°C.

8.6. Disassembling NHVf pump with mechanical sealing

Refer to **Fig. 12.2** for disassembly of the pump with mechanical seal.

After separating the motor from the pump and transporting to workstation start the dismantle:

- Remove nuts (37) fixing spiral casing (219) and sealing casing (76).
- Remove spiral casing (219).
- Remove nut (39) securing the impeller (1).
- Slide the impeller (1) from the shaft (225) and remove key (44).
- Remove distance sleeve (73) and mechanical sealing (72).
- Remove nuts (38) connecting bearing casing (221) with sealing casing (76).
- Remove sealing casing (76).
- Remove bearing covers (222), (223) and retaining rings (228) using the pliers.
- Remove the shaft (225) with bearings (227) from bearing casing (221).
- Remove the bearings (227) from shaft (225) with puller

Table 8.2 Mechanical seal dimensions

Size group	Shaft diameter, mm	Mechanical seal diameter, mm
<i>A</i>	24	30
<i>B</i>	32	40
<i>C</i>	42	50
<i>D</i>	48	60

8.7. Mounting of bearings

All removed parts must be cleaned and checked for excessive wear. Replace damaged or worn parts with new genuine parts. Clean all sealing surfaces.

- Push bearings (227) on the shaft (225).
- Initially mounted shaft (225) insert in bearing casing (221).
- Mount the bearing covers (228) and fixing rings using pliers (222) (223).

8.8. Mounting of the mechanical sealing

- Install fixed ring (72) with O-ring , in sealing casing (76).
- Insert O-ring (59) into sealing casing (76).
- Put sealing casing (76) on the shaft (225).
- Tighten sealing casing (76) to bearing casing (221).
- Install the moving part of the mechanical sealing (72) on the shaft (225) and place it in sealing casing (76).
- Embed distance sleeve (73) on the shaft (225).

8.9. Mounting of cord sealing

- Install the gland seal (217) on the shaft (225).
- Place cord (205) and spacer ring (207) in cord sealing casing (206).
- Insert another cords (205). Each next cord should be moved relative to previous by 90°.
- Sealing casing (208) along with cords (205) and spacer ring (207) put on the shaft (225) and tighten the gland (217) to cord sealing casing (206) using the nuts (212).

8.10. Mounting of impeller

- Insert key (44).
- Install impeller (1) on shaft (225) and secure with nut (39).
- Put ring type „O” (59) on seal casing (208).
- Install spiral casing (219) and fasten with nuts (37) to seal casing (208).
- Tighten gland seal (217) to seal casing (208) using the nuts (212).

8.11. Recommended spare parts

Table 8.3 Recommended spare parts

Part name	Number of system parts						
	2	3	4	5	6-7	8-9	10+
Shaft	1	1	2	2	2	3	30%
Impeller	1	1	1	2	2	3	30%
Gasket ring	2	2	2	4	4	6	50%
Ball bearing	2	2	4	4	6	8	50%
Bearing casing	-	-	-	-	-	1	2 pcs.
O-ring	4	6	8	8	9	12	150%
Cord seal	4	4	6	6	6	8	40%
Seal sleeve	1	1	1	2	2	2	20%

9. TROUBLESHOOTING

Any repairs on the pump unit within the warranty period without the manufacturer approval will void the warranty.

Before troubleshooting, check if the measuring instruments are reliable and precise. Shut down the pump before troubleshooting and do it according to a manual.

Problem	Cause	Solution
1	2	3
Pump rotates but does not transfer medium	a) water level below strainer b) pump not primed (check valve leakage, air not released, pump not primed) c) valve between tank and pump closed d) suction line leakage e) strainer clogged f) pump damaged g) pump shaft leakage at the mechanical seal	a) check water level and/or lower strainer b) check valve, release air and prime pump c) check if valve is open d) check suction line and connectors e) clean strainer f) remove and repair the pump unit g) replace packing

Motor overload	a) motor power too low or pump delivery too high b) medium density or viscosity too high c) mechanical losses too high due to wear	a) check pump compatibility b) check pump compatibility, c) remove and repair the pump unit
Pump capacity and pressure too low	a) suction and discharge line diameter too small b) incorrect pump direction c) impeller mouth ring worn d) air bubbles in the suction and discharge line e) discharge line gate valve jammed f) automatic discharge line gate valve closing g) impeller ducts clogged h) motor speed too low i) error in delivery head calculations	a) check pump compatibility b) reverse motor direction by replacing two phase conductors c) replace rings or impeller d) verify suction line condition and medium level e) excessive gate valve play, check gate valve f) excessive gate valve play, check play g) clean impeller, check operating conditions h) check frequency and supply voltage i) check all calculations
Unit vibrations and water hammers	a) pump capacity higher than specified in the order b) worn bearings and mouth rings c) solids in impeller ducts d) impeller worn or damaged e) incorrect motor direction	a) check pump compatibility b) replace bearings, mouth rings or impeller c) clean impeller, check operating conditions d) replace impeller, incorrect operating conditions e) reverse motor direction by replacing two phase conductors

10. NHVf PUMP WEIGHT AND DIMENSIONS

Pump type	Size group	Shaft diameter/length, mm	Weight, kg
32-125	A	24/360	30
32-160			31
32-200			41
32-250			50
40-125			32
40-160			33
40-200			45
40-250			50
50-125			35
50-160			36
50-200			45
50-250			51
65-125			42
65-160			44
65-200			50
80-160			52
40-315	B	32/470	84
50-315			87
65-250			77
65-315			93
65-400			109
80-200			70
80-250			87
80-315			100
100-160			91
100-200			84
100-250			92
100-315			110
125-200			108
125-250			121
150-200			166
150-250			161

80-400	C	42/530	148
100-400			163
125-315			152
125-400			175
150-315			178
150-400			203
200-260			255
200-315	D	48/670	263
200-400			350
200-500			400

11. RECYCLING

This product and its parts must be recycled in line with applicable environmental protection regulations. To this end, employ local, public or private companies dealing with waste and reusable material recycling.

If it is not possible, contact the head office or a local service point of Hydro-Vacuum S.A.

12. SUPPLEMENT A. ASSEMBLY DRAWINGS

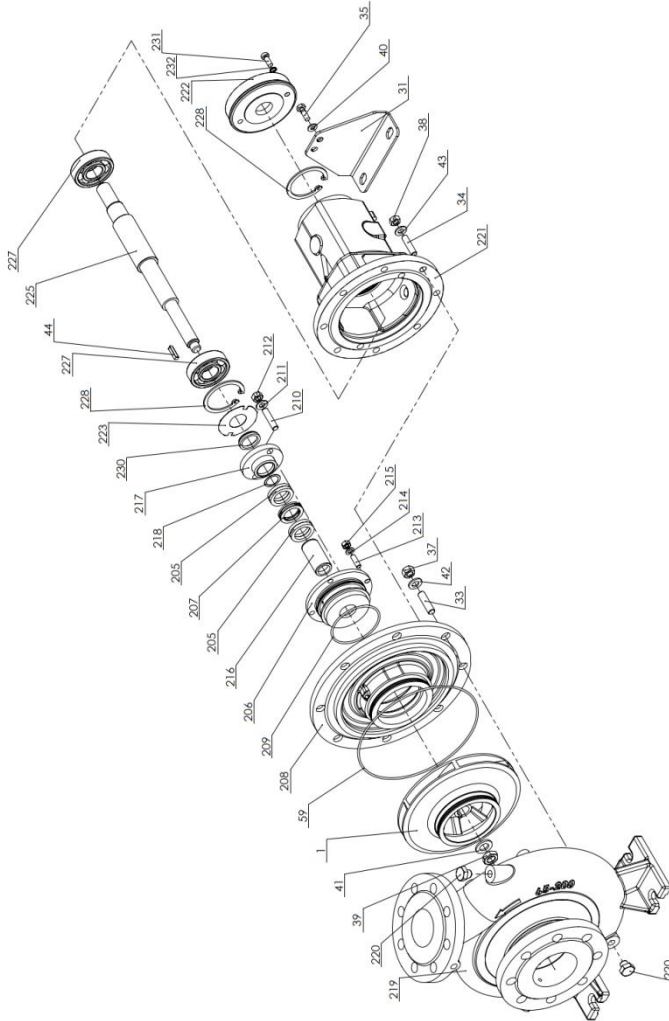


Fig. 12.1 NHVf pump assembly drawing with cord sealing

- 1 - impeller; 31 - foot; 33,34,210,213 - stud bolt; 35,231 - bolt; 37,38,39,212,215 - nut;
 40,41,42,43,232,211,214 - washer; 44 - key; 59,209 - O-ring;
 205 - cord; 206 - cord sealing sleeve; 207 - spacer ring;
 208 - sealing casing; 216 - shaft protection sleeve; 217 - gland seal; 218 - seal;
 219 - pump casing; 220 - square head bolts with collar; 221 - bearing casing;
 222,223 - bearing cover; 225 - shaft; 227 - ball bearing;
 228 - circlip; 230 - ACA ring;

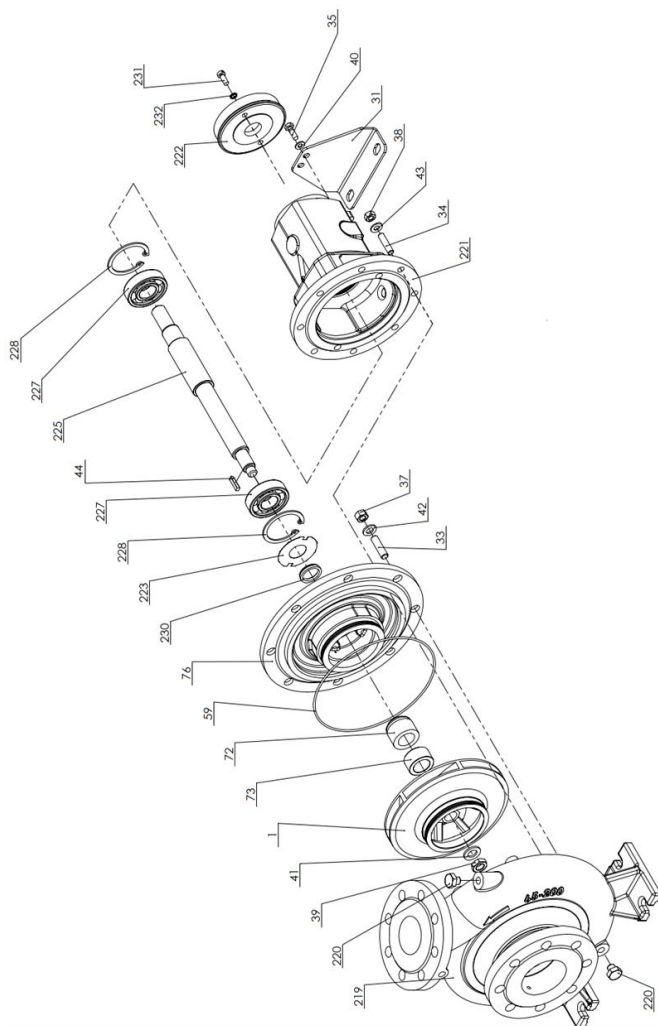


Fig. 12.2 NHVf pump assembly drawing with mechanical sealing

1 - impeller; 31 - foot; 33,34 - stud bolt; 35,231 - bolt; 37,38,39 - nut;
 40,41,42,43,232 - washer; 44 - key; 59 - O-ring; 72 - mechanical seal; 73 - distance sleeve; 76 -
 sealing casing; 219 - pump casing; 220 - Square head bolts with collar; 221 - bearing casing;
 222,223 - bearing cover; 225 - shaft; 227 - ball bearing; 228 - circlip; 230 - ACA ring;



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